

Name: _____

Date: _____

College Algebra 1220 – Exam 1 Practice

Summer 2026

Instructions: Answer all questions. Show all your work for full credit. No calculators or cell phones are allowed during the exam.

1. Simplify:

a) $(2x^3y^2)^2$

b) $16^{\frac{3}{2}}$

c) $(\frac{3}{2})^2 \cdot 5 \cdot 2^{-2}$

2. Solve for x : $|2x - 5| = 7$

3. Combine the rational expressions:

a) $\frac{2}{x} + \frac{3}{x+1}$

b) $\frac{\frac{2}{x} + 1}{\frac{3}{2x}}$

4. Solve the radical equation: $\sqrt{2x+3} = x$

5. Let $f(x) = x^2 - 3x + 2$.

(a) Find the x-intercept of the function.

b) Evaluate $f(x+h)$.

6. Let $g(x) = \frac{2x + 1}{\sqrt{x - 3}}$. Answer the following question:

a) The domain of $g(x)$.

b) The x -intercept(s).

c) The y -intercept.

7. Find the equation of a line perpendicular to $2x - 3y = 7$ and passing through $(6, -1)$.

8. **Application:** A taxi company charges a flat fee of \$4 plus \$2 per mile.

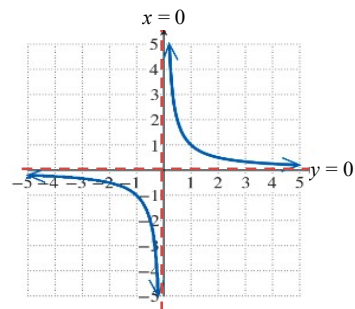
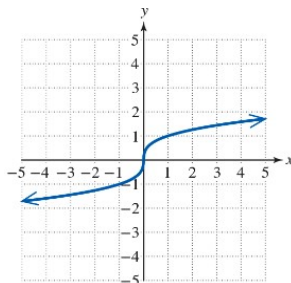
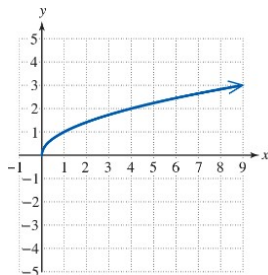
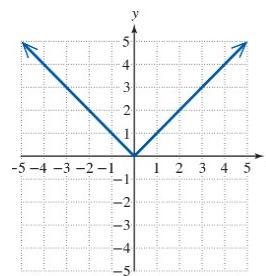
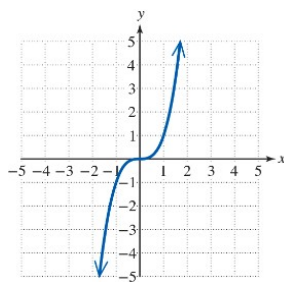
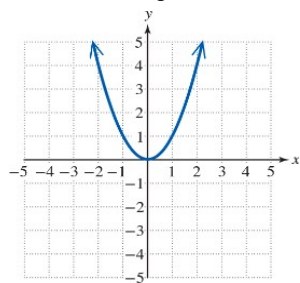
(a) Write a linear equation that models the total cost C in terms of miles m .

(b) How much does a 10-mile trip cost?

(c) How far can you travel with \$24?

Unit 6 Review

1. Write the equation of each function.



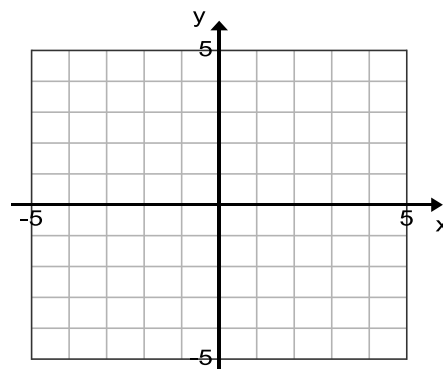
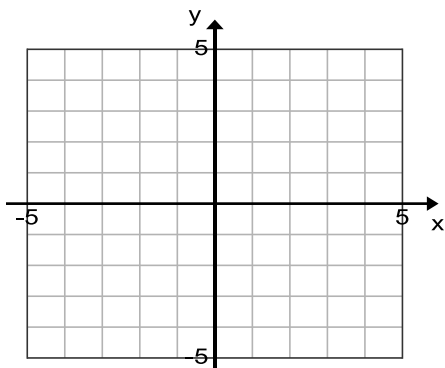
2. Graph each function by graphing transformations in the indicated stages. Clearly indicate 3 points on each curve by putting in an appropriate dot. **Note:** The first set of axes should have 3 clearly drawn curves, **each one labeled**. What you are graphing should be put into the “y=”. The 2nd set of axes should have only your final curve.

(a) $f(x) = 3 - 2(x + 1)^2$

Basic curve Stretch/Shrink Reflection: _____

Shift(s): _____

y = _____ y = _____ y = _____

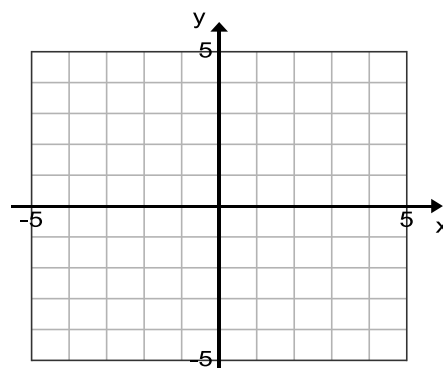
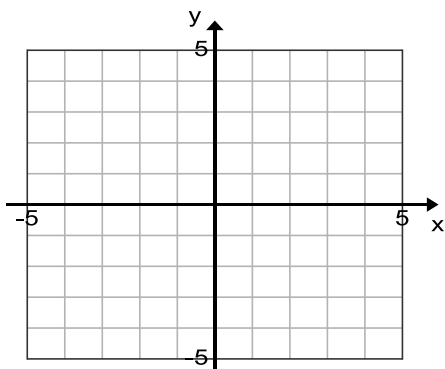


(b) $g(x) = \sqrt{2 - x} - 3$

Basic curve Stretch/Shrink Reflection: _____

Shift(s): _____

y = _____ y = _____ y = _____



3. Let $f(x) = \frac{3}{x+2}$. Find the ARC of f over the interval $[1, 7]$. Use proper function notation.

4. Consider the function $f(x) = \begin{cases} -x^2 + 3, & x < 1 \\ 2x + 1, & 1 \leq x < 3 \\ 4, & x > 3 \end{cases}$

Graph the function.

(a) evaluate:

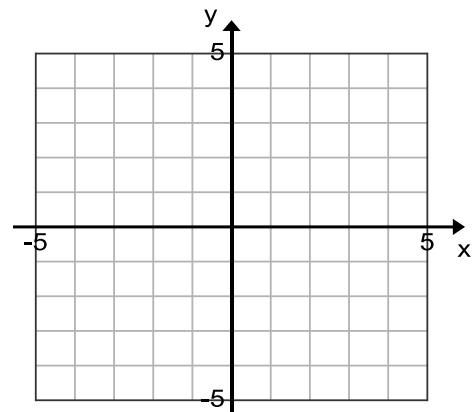
$$f(3) =$$

$$f(-5) =$$

$$f(0) =$$

$$f(1) =$$

(b)



5. Let $f(x) = x^2 - x + 1$ and $g(x) = \frac{3}{x}$ Evaluate each difference quotient.

a) $\frac{f(x+3) - f(x)}{h}$

b) $\frac{g(x+h) - g(x)}{h}$

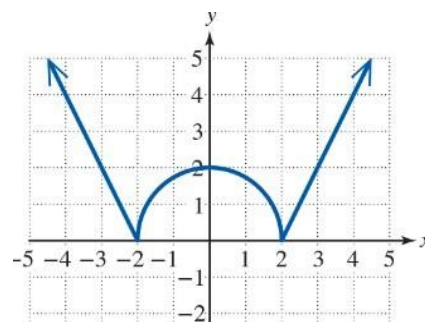
6. Use interval notation to write the interval(s) over which $f(x)$ is increasing or decreasing. Identify the value and location of any relative maxima or minima of the function.

Increasing: _____

Decreasing: _____

Relative minima:

Relative maxima:



7. Let $f(x) = \frac{x+2}{x+1}$ and $g(x) = \frac{1}{x(x+1)}$. Find

(a) $(f + g)(x)$ and its domain

(b) $(\frac{f}{g})(x)$ and its domain.

8. Let $f(x) = \frac{5}{x^2-1}$, $g(x) = \sqrt{x+1}$ and $h(x) = \frac{1}{x}$. Find the following.

(a) $(f \circ g)(x)$ and its domain

(b) $(f \circ h)(x)$ and its domain

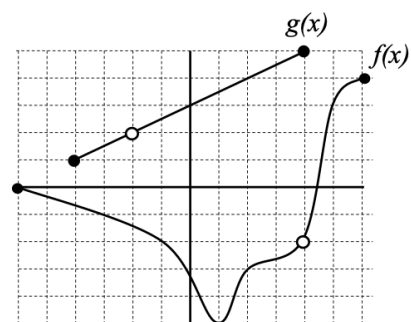
9. Let f and g be the functions shown in the graph. Find

(a) $(f + g)(2)$

(b) $(g \circ f)(6)$

(c) $(f \circ g)(-1)$

(d) $g^{-1}(1)$



10. Let $f(x) = \frac{1-2x}{5x+6}$. Find $f^{-1}(x)$ and state its range.

11. Let $f(x) = (x + 2)^2 - 1$, $x \geq -2$. Find $f^{-1}(x)$ and graph both on the same set of axes.

